# ZULHELMI SYAHTIAR

**Fresh Graduate D3 Industrial Electronics — IoT, Microcontrollers & Control Systems**

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Portfolio | LinkedIn

## SUMMARY

Fresh graduate in Industrial Electronics from Politeknik Negeri Jakarta with internship experience in power system protection at PT PLN. Strongly interested in the Internet of Things (IoT), microcontrollers, programming, and control systems, with hands-on experience across multiple microcontroller-based projects and simple automation system development. Eager to continuously learn, innovate, and apply modern technologies in industrial environments.

## EXPERIENCE

**PT PLN (Persero) ULTG Petukangan – South Tangerang** *Jul 2025 – Oct 2025*
*Protection Maintenance Intern (HAR PRO)*

- Assisted with periodic maintenance and monitoring of substation protection systems.
- Performed testing of protection relays: OCR, GFR, Differential, Distance Relay, OVR/UVR, Buchholz, and Thermal Overload Relay.
- Involved in operational maintenance and system testing during DSS migration at a 150kV substation.
- Took part in Battery Capacity Tests (BCT) to maintain the reliability of substation backup power.
- Assisted with non-cascade testing on transformer incoming panels.

**Independent Clients – Jabodetabek** *Mar 2023 – Present*
*Freelance Graphic Designer & Photographer*

- Provided freelance event photography, banner design, and social media graphic content for clients and student organizations.

## EDUCATION

**Politeknik Negeri Jakarta – Depok, Jawa Barat** *2023 – 2026*
*D3 Industrial Electronics — GPA 3.55/4.00*

- Final Project: "Dual DC Motor Control Trainer Kit – PID – Fuzzy Control Based on ESP32 with 3D Digital Twin Visualization".
- Sub-title: "Design of Web Monitoring and 3D Digital Twin Visualization Based on Real-Time Database for the Trainer Kit".
- Teaching Assistant for a Community Service project installing a solar-powered Hydroponic Green House.

**SMAN 64 Jakarta – Jakarta Timur** *2019 – 2022*
*High School – Science Major (MIPA)*

## FEATURED PROJECTS

- DC Motor Control Trainer Kit (2025–2026): Web monitoring dashboard (JavaScript ES6, HTML5, Tailwind CSS v4) and an interactive 3D Digital Twin (Three.js r160) synchronized in real time with ESP32 hardware via Firebase Realtime Database.
- IoT Web Monitoring Dashboard (2025): Monitoring of temperature and soil & air humidity based on a MySQL database.
- Plant Monitoring Robot (2025): ESP32-C3 with DHT11 and soil moisture sensors, IoT-based.
- Object and Color Detection Mobile Robot (2026): Developed a real-time object and color detection system using a Raspberry Pi-based camera.
- Logic Gate Training Module (2024): Practice kit for AND, OR, NOT, NAND, NOR, XOR, Full Adder, Multiplexer, Demultiplexer, Flip-flop, and BCD-to-7-Segment Decoder.
- Line Follower & Obstacle-Avoiding Robots (2025): Built an OP-Amp-based line follower robot and an Arduino Uno-based obstacle-avoiding robot.
- Football Scoreboard (2024): Designed a two-team scoreboard using seven-segment displays.
- Automatic Hand Washing Machine (2024): Touchless system based on Arduino Uno and an infrared sensor to automatically dispense soap/water.
- 9V Power Supply based on IC 7809 (2024): Design, assembly, and testing of a stable power supply for small-scale electronic devices.
- Python Snake Game (2024): Built a simple Snake game in Python, controlled with a joystick module integrated via Arduino Uno.
- Numa Space (2026): A virtual study room and real-time study-buddy matchmaking platform, with P2P video/audio interaction via WebRTC, Socket.IO chat, a private room access code system, and Pomodoro Timer integration.

## ORGANIZATIONAL EXPERIENCE

**PSYCHOROBOTIC PNJ – Depok, Jawa Barat** *Mar 2025 – Dec 2025*

*Communication & Information Department*

Student Study Group (KSM) at Politeknik Negeri Jakarta focused on Robotics and IoT.

- Created all design assets needed by the organization.
- Documented all organizational activities.
- Managed the organization's social media.
- Created educational content about robotics.
- Took part in robotics competitions.

## CERTIFICATIONS

- **BNSP Competency Certification – Instrumentation Technician — BNSP (2026)**

Competent. Competency test: zero-span configuration and scaling of a level transmitter (4–20 mA) on a PLC, including linearity and measurement-deviation analysis.

- **TOEIC Prediction Test — Politeknik Negeri Jakarta (2026)**

Score 620/990 (Listening 260, Reading 360).

## SKILLS

- **IoT & Microcontrollers:** ESP32, Arduino, Raspberry Pi, Firebase (Realtime Database, Firestore & Hosting), Computer Vision (OpenCV)
- **Control & Industrial Automation:** PID Controller, Fuzzy Logic Controller, PLC programming, SCADA (Schneider Electric, Vijeo Citect)
- **Protection & Testing Systems:** Protection Relays (OCR, Differential, REF MiCOM), OMICRON TestUniverse, GOOSE Configuration
- **Electronics Practical Skills:** Soldering, Desoldering, Wiring, Use of Measurement Tools (Multimeter, Oscilloscope, etc.)
- **Programming:** Python, C/C++, HTML, CSS, JavaScript, TypeScript, React, Node.js, Express
- **Design & Simulation:** NI Multisim, Proteus, AutoCAD, Fusion 360, Eagle
- **Visualization & Real-time:** Three.js, WebRTC, Socket.IO
- **Databases:** MySQL, PostgreSQL (Supabase)
- **Microsoft Office:** Word, Excel, PowerPoint
- **Languages:** Indonesian (native), Malay, English (TOEIC Prediction 620/990 – working proficiency)